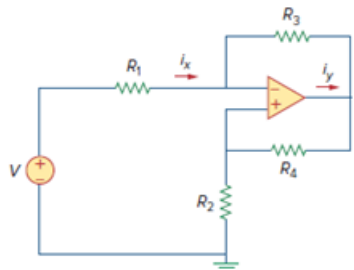


Problem

Using Fig. 5.55, design a problem to help students better understand inverting op amps.

Figure. 5.55:



Step-by-step solution

Step 1 of 8

Refer to Figure 5.55 in the textbook.

Represent the node voltages and currents in the circuit.

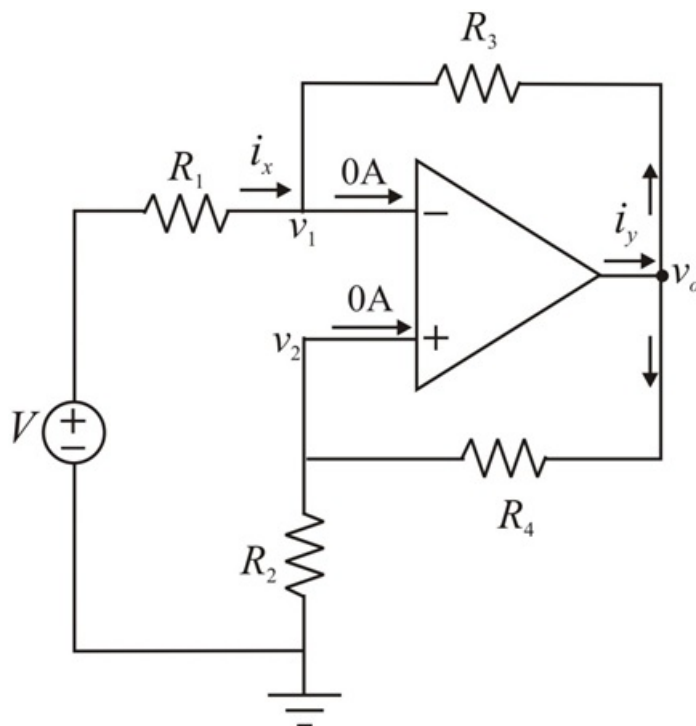


Figure 1

Step 2 of 8

Consider that the voltages at inverting and non-inverting terminals of an ideal op amp are equal due to virtual ground.

$$v_1 = v_2$$

Apply Kirchhoff's current law at node v_1 .

$$\frac{v_1 - V}{R_1} + \frac{v_1 - v_o}{R_3} = 0$$

$$\frac{R_3(v_1 - V) + R_1(v_1 - v_o)}{R_1 R_3} = 0$$

$$R_3(v_1 - V) + R_1(v_1 - v_o) = 0$$

$$R_3 v_1 - R_3 V + R_1 v_1 - R_1 v_o = 0$$

$$(R_1 + R_3)v_1 - R_3 V - R_1 v_o = 0$$

$$(R_1 + R_3)v_1 = R_3 V + R_1 v_o$$

$$v_1 = \frac{R_3 V + R_1 v_o}{R_1 + R_3}$$

Step 3 of 8

Apply Kirchhoff's current law at node v_2 .

$$\frac{v_2}{R_2} + \frac{v_2 - v_o}{R_4} = 0$$

$$\frac{R_4 v_2 + R_2(v_2 - v_o)}{R_2 R_4} = 0$$

$$R_4 v_2 + R_2(v_2 - v_o) = 0$$

$$R_4 v_2 + R_2 v_2 - R_2 v_o = 0$$

$$(R_2 + R_4)v_2 - R_2 v_o = 0$$

$$(R_2 + R_4)v_1 - R_2 v_o = 0 \text{ (since, } v_1 = v_2 \text{)}$$

Step 4 of 8

Substitute the expression of v_1 in the equation.

$$(R_2 + R_4) \left(\frac{R_3 V + R_1 v_o}{R_1 + R_3} \right) - R_2 v_o = 0$$

$$\frac{(R_2 + R_4)(R_3 V + R_1 v_o)}{R_1 + R_3} - R_2 v_o = 0$$

$$\frac{(R_2 + R_4)R_3 V + (R_2 + R_4)R_1 v_o - (R_1 + R_3)R_2 v_o}{R_1 + R_3} = 0$$

$$(R_2 + R_4)R_3 V + (R_2 + R_4)R_1 v_o - (R_1 + R_3)R_2 v_o = 0$$

$$(R_2 + R_4)R_3 V + R_2 R_1 v_o + R_4 R_1 v_o - R_1 R_2 v_o - R_3 R_2 v_o = 0$$

$$(R_2 + R_4)R_3 V + R_4 R_1 v_o - R_3 R_2 v_o = 0$$

$$(R_2 + R_4)R_3 V - (R_2 R_3 - R_1 R_4)v_o = 0$$

$$(R_2 R_3 - R_1 R_4)v_o = (R_2 + R_4)R_3 V$$

$$v_o = \frac{(R_2 + R_4)R_3 V}{R_2 R_3 - R_1 R_4}$$

Therefore, the output voltage is $v_o = \frac{(R_2 + R_4)R_3 V}{R_2 R_3 - R_1 R_4}$.

Step 5 of 8

Assume some values for the resistances and the input voltage to find the value of output voltage.

$$R_1 = 1 \text{ k}\Omega$$

$$R_2 = 2 \text{ k}\Omega$$

$$R_3 = 3 \text{ k}\Omega$$

$$R_4 = 4 \text{ k}\Omega$$

$$V = 5 \text{ V}$$

Substitute these values in the expression of output voltage.

$$\begin{aligned} v_o &= \frac{(2 \text{ k}\Omega + 4 \text{ k}\Omega)(3 \text{ k}\Omega)(5 \text{ V})}{(2 \text{ k}\Omega)(3 \text{ k}\Omega) - (1 \text{ k}\Omega)(4 \text{ k}\Omega)} \\ &= \frac{(6)(3)(5 \text{ V})}{6 - 4} \\ &= 45 \text{ V} \end{aligned}$$

Thus, the value of output voltage is, $\boxed{45 \text{ V}}$.

Step 6 of 8

Find the value of current i_x .

$$i_x = \frac{V - v_1}{R_1}$$

Substitute the expression of v_1 in the equation.

$$\begin{aligned} i_x &= \frac{V - \left(\frac{R_3 V + R_1 v_o}{R_1 + R_3} \right)}{R_1} \\ &= \frac{1}{R_1} \left(\frac{R_1 V + R_3 V - R_3 V - R_1 v_o}{R_1 + R_3} \right) \\ &= \frac{R_1 (V - v_o)}{R_1 (R_1 + R_3)} \\ &= \frac{V - v_o}{R_1 + R_3} \end{aligned}$$

Substitute the component values to find the value of i_x .

$$\begin{aligned} i_x &= \frac{5 - 45}{1 \text{ k}\Omega + 3 \text{ k}\Omega} \\ &= \frac{-40}{4 \text{ k}\Omega} \\ &= -10 \text{ mA} \end{aligned}$$

Thus, the value of current i_x is $\boxed{-10 \text{ mA}}$.

Step 7 of 8

Apply Kirchhoff's current law at output terminal.

$$\begin{aligned} i_y &= \frac{v_o - v_1}{R_3} + \frac{v_o - v_2}{R_4} \\ &= \frac{v_o - v_1}{R_3} + \frac{v_o - v_1}{R_4} \quad (\text{since, } v_1 = v_2) \\ &= (v_o - v_1) \left(\frac{1}{R_3} + \frac{1}{R_4} \right) \end{aligned}$$

Substitute the expression of v_1 in the equation.

$$\begin{aligned} i_y &= \left[v_o - \left(\frac{R_3 V + R_1 v_o}{R_1 + R_3} \right) \right] \left(\frac{1}{R_3} + \frac{1}{R_4} \right) \\ &= \left(\frac{R_1 v_o + R_3 v_o - R_3 V - R_1 v_o}{R_1 + R_3} \right) \left(\frac{R_3 + R_4}{R_3 R_4} \right) \\ &= \left[\frac{R_3 (v_o - V)}{R_1 + R_3} \right] \left(\frac{R_3 + R_4}{R_3 R_4} \right) \\ &= (v_o - V) \left[\frac{R_3 + R_4}{R_4 (R_1 + R_3)} \right] \end{aligned}$$

Step 8 of 8

Substitute the component values to find the value of i_y .

$$\begin{aligned} i_y &= (45 - 5) \left[\frac{3 \text{ k}\Omega + 4 \text{ k}\Omega}{(4 \text{ k}\Omega)(1 \text{ k}\Omega + 3 \text{ k}\Omega)} \right] \\ &= \frac{40(7 \text{ k}\Omega)}{(4 \text{ k}\Omega)(4 \text{ k}\Omega)} \\ &= 40(437.5 \times 10^{-6}) \\ &= 17.5 \text{ mA} \end{aligned}$$

Thus, the value of current i_y is 17.5 mA.